

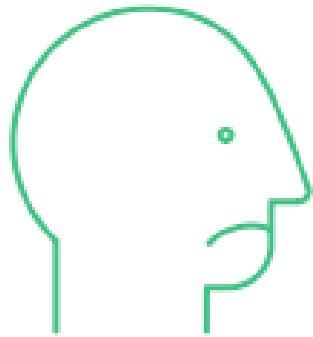
Team Members



- ✓ Rewida Ramadan
- ✓ Hend Mohammed
- ✓ Mayer Salah
- ✓ Nada Ragab
- ✓ Alaa Adel

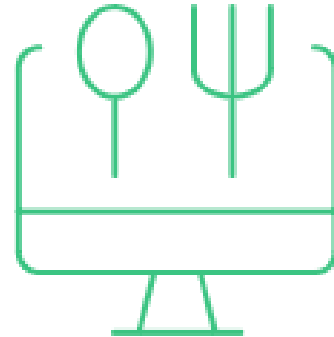
Global Food & Nutrition

Project Idea



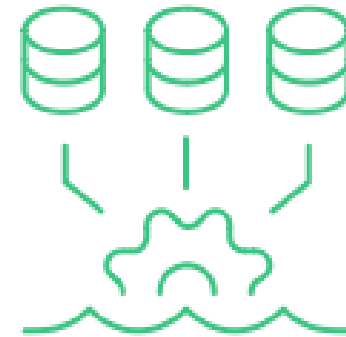
Problem

People struggle with food choices due to unclear information and risks.



Solution

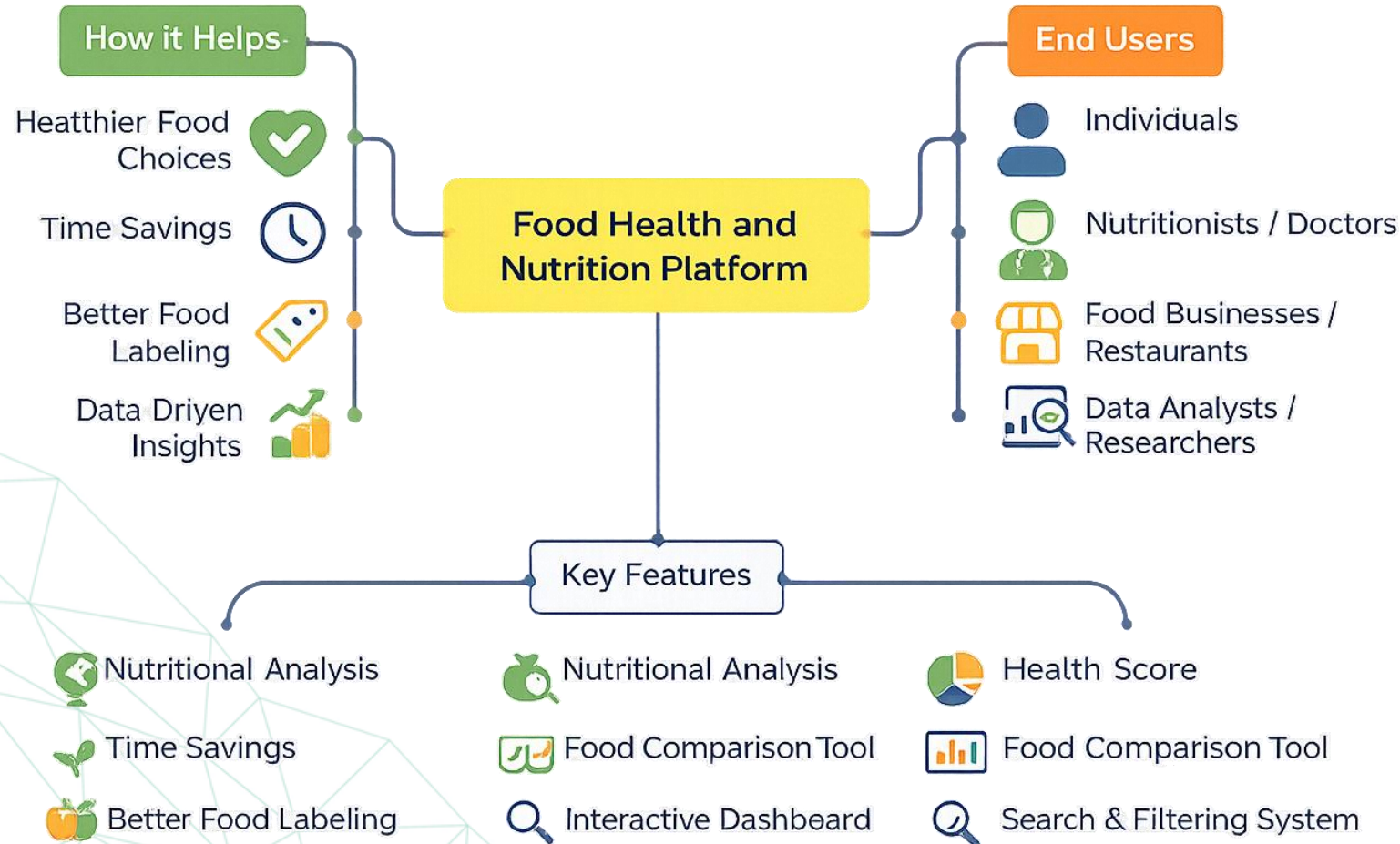
Cleaned data, Power BI dashboards, and integrated insights are provided.



Unique Value

Combines data engineering and business intelligence for healthier choices.

End Users + Features

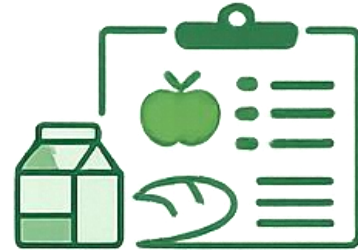


Data Structure



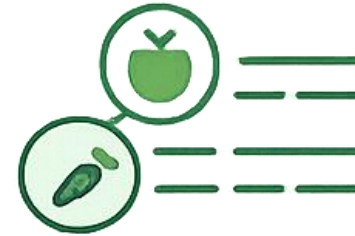
Architecture

CSV dataset
processed using
Pytlion and visualized
in-Power BI



Key Entities

Food products,
nutritional-values,
catteggies, brands,
ingredients, and
aliergens



Relationships

Food products have
nutritional values,
belong to categories,
and contain ingredients.



Data Flow

Data collection from
Kaggies cleaning
with Python, and
visualization in Power BI

Programming Languages + Frameworks

Programming Language and Tools



Pvthon



Power BI



CSV Files



Kaggle Dataset



Random Forest
Regression Modal



Random Forest
Regression Model

Frameworks / ML Libraries



Pandas



NumPy



Matplotlib & Seaborn



Scikit-learn



Random Forest
Regression Model

Technologies



Power BI
(Visualization)



Machine Learning
Techniques



Data Preprocessing
& Standardization



Medallion
Architecture

Live Application + Test

Application Status



✓ Interactive Power BI dashboard



✓ Machine Learning
(Random Forest
Regression Model)

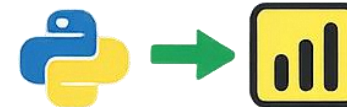
✓ Insights & Predictions
(Model Evaluation)

✓ Insights & Predictions
(Model Evaluation)

Testing Phases



Data validation
(cleaning & accuracy)



Integration testing
(Python → Power BI)

Deliverables



REPORTS & DOCUMENTATION

- ✓ Technical Documentation
- ✓ Data Analysis Report
- ✓ Dashboard Explanation

TIMELINE



- ✓ Data Collection & Cleaning
- ✓ Analysis & Modeling
- ✓ Dashboard Development
- ✓ Final Delivery

FINAL OUTPUTS



- ✓ Interactive Power BI dashboard
- ✓ Machine Learning Model (Random Forest)
- ✓ Cleaned Dataset (CSV)
- ✓ Source Code (Python)

Deliverables

Build a ML Model



Predict food health score
using nutritional values



Improve food quality
analysis and decision-making



Support healthier
food selection



Generate data-driven
nutritional insights

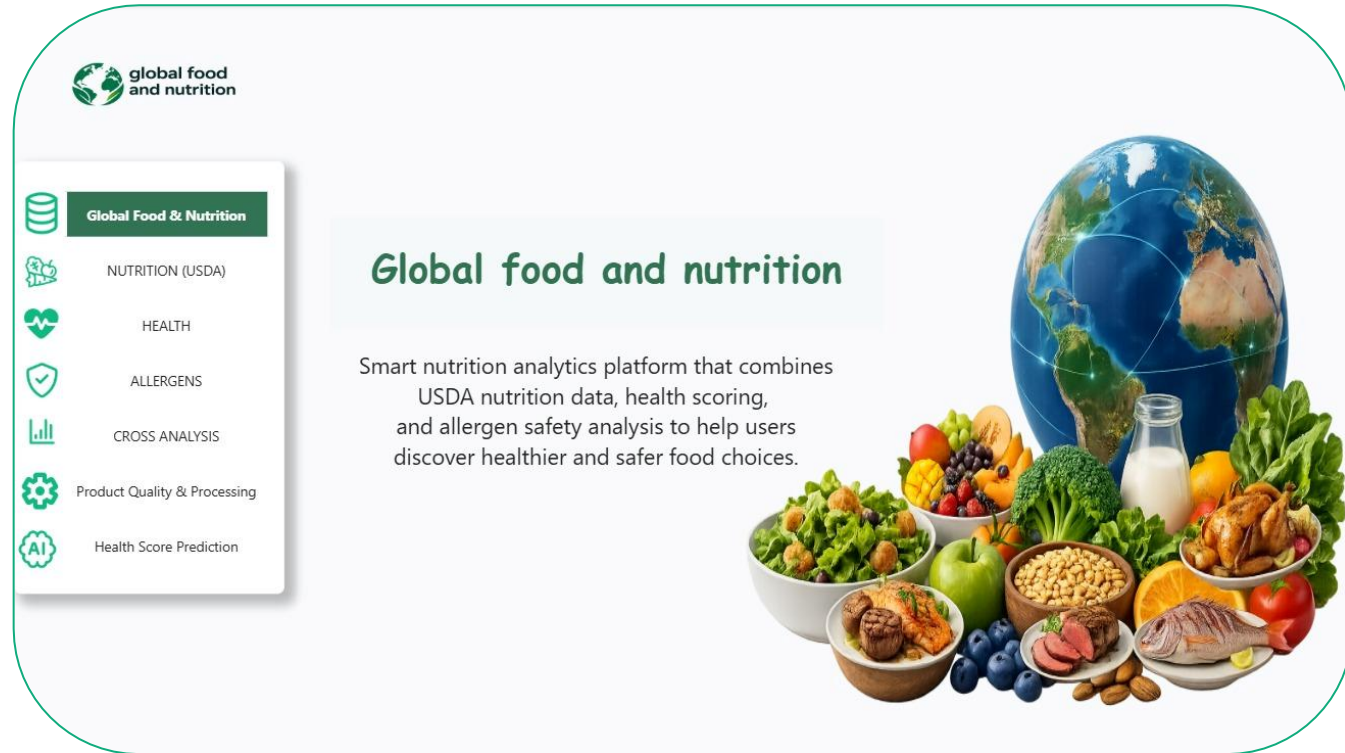


- Calories
- Sugar
- Fat
- Protein
- Sodium

Analyze calories, sugar,
fat, protein, and sodium

Objectives of the project

- Analyze nutritional values of food products.
- Detect allergens and improve food safety.
- Use Machine Learning to predict health scores.
- Evaluate food health and quality.

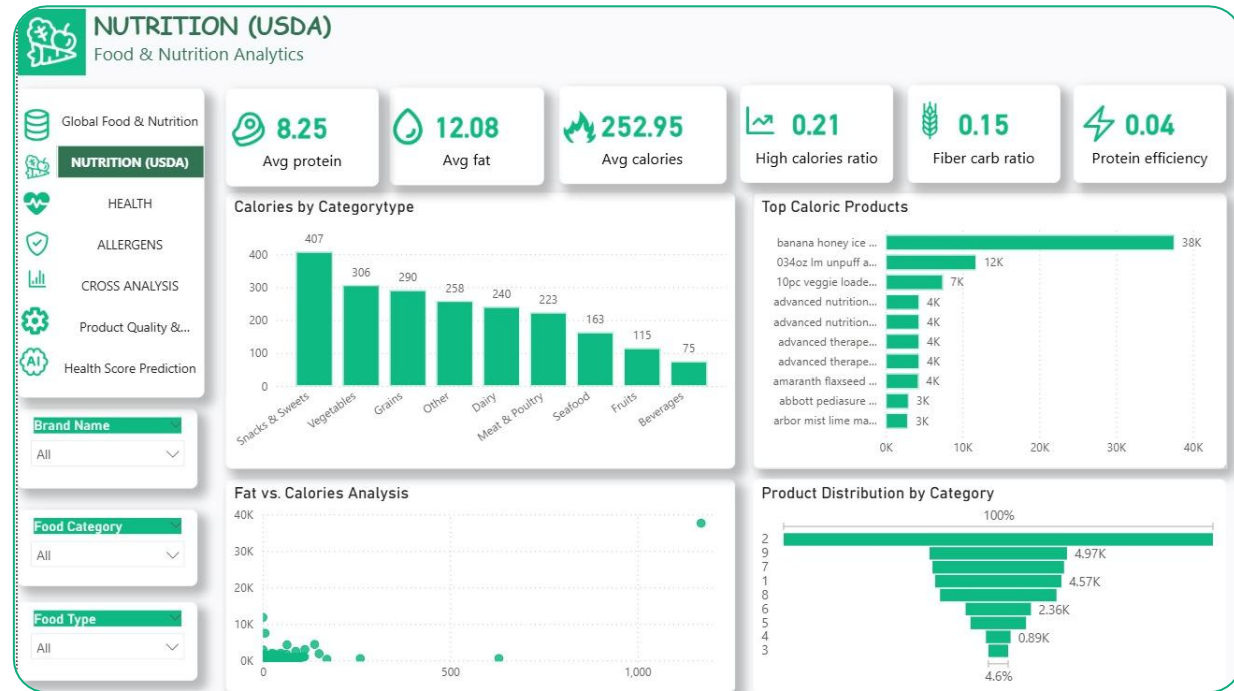
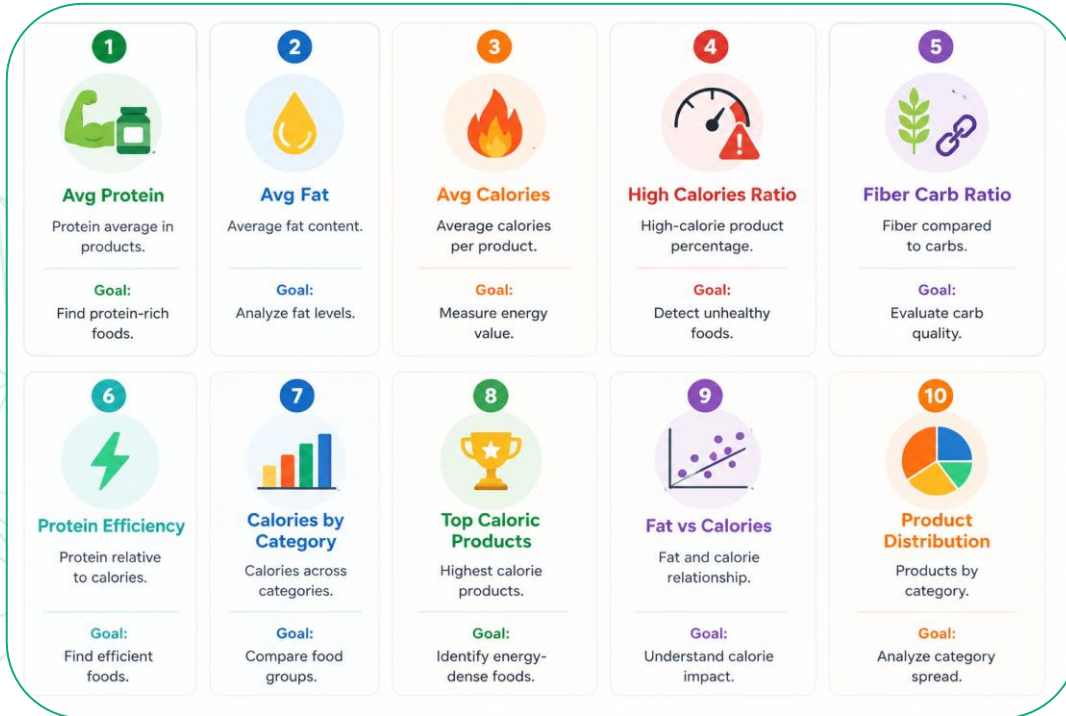


The image shows a digital interface for the 'Global Food & Nutrition' platform. On the left, a sidebar lists various features: 'Global Food & Nutrition' (with a database icon), 'NUTRITION (USDA)' (with a fork and knife icon), 'HEALTH' (with a heart icon), 'ALLERGENS' (with a shield icon), 'CROSS ANALYSIS' (with a bar chart icon), 'Product Quality & Processing' (with a gear icon), and 'Health Score Prediction' (with an AI icon). The main content area has the title 'Global food and nutrition' and a description: 'Smart nutrition analytics platform that combines USDA nutrition data, health scoring, and allergen safety analysis to help users discover healthier and safer food choices.' To the right of the text is a collage of various food items including fruits, vegetables, grains, and proteins, with a globe in the background.



Nutrition (USDA)

Analyze food and nutrition data to evaluate product health products.





Health

Evaluate product health by analyzing health scores, calories, sugar, and sodium to support healthier food decisions



Low Health

Low-health product ratio.

Goal: Detect unhealthy foods.



Avg Health Score

Average health rating.

Goal: Measure food health quality.



High Health

Healthy product percentage.

Goal: Identify healthy foods.



Health Performance Index

Overall health indicator.

Goal: Evaluate health performance.



Active Health Tiers

Number of health levels.

Goal: Classify products by health.



Health Score Distribution

Distribution of health scores.

Goal: Analyze health balance.



Category Calorie Trends

Calories across categories.

Goal: Compare category energy levels.



Sugar Content by Category

Sugar levels per category.

Goal: Detect high-sugar foods.



Sodium & Sugar View

Sodium and sugar relationship.

Goal: Analyze health risks.



Nutrition & Health

Food & Nutrition Analytics

- Global Food & Nutrition
- NUTRITION (USDA)
- HEALTH
- ALLERGENS
- CROSS ANALYSIS
- Product Quality & Processing
- Health Score Prediction

Food Type
All

Health Score
60 75

0.25

Low health

67.50

Avg health score

0.50

High health

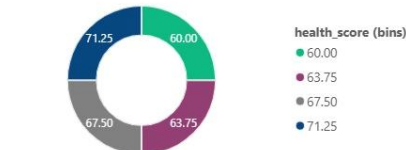
0.26

Health Performance Index

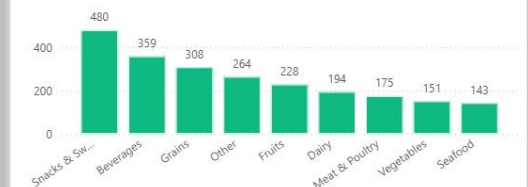
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Active Health Tiers

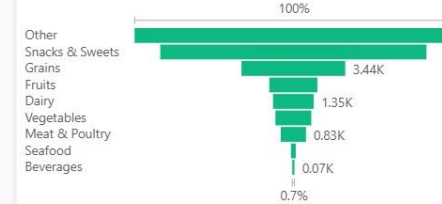
Health Score Distribution



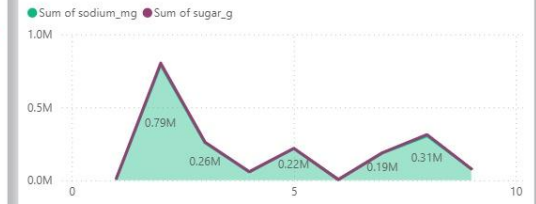
Category Calorie Trends



Sugar Content by Category



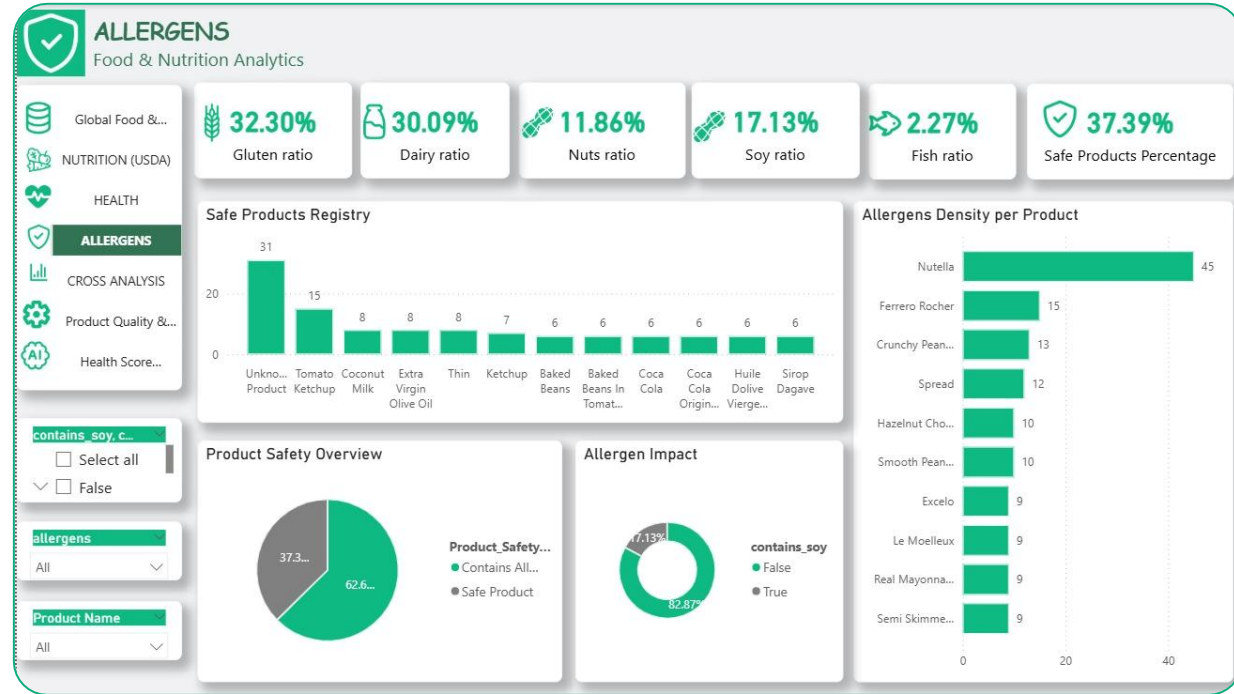
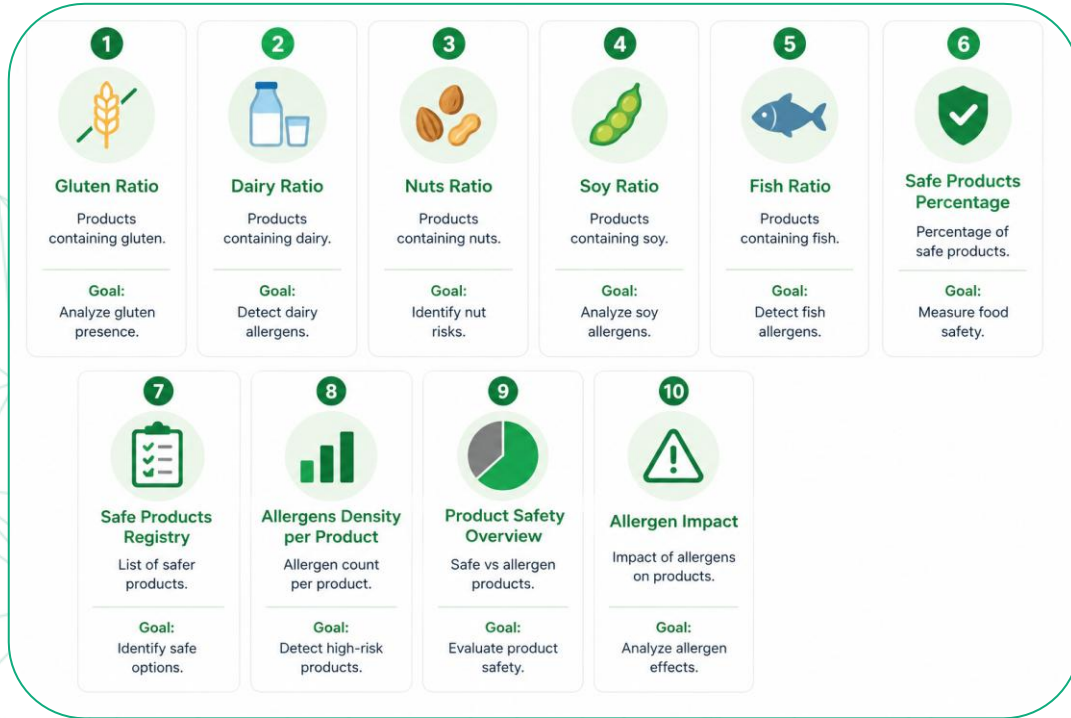
Sodium & Sugar Cumulative View





Allergens

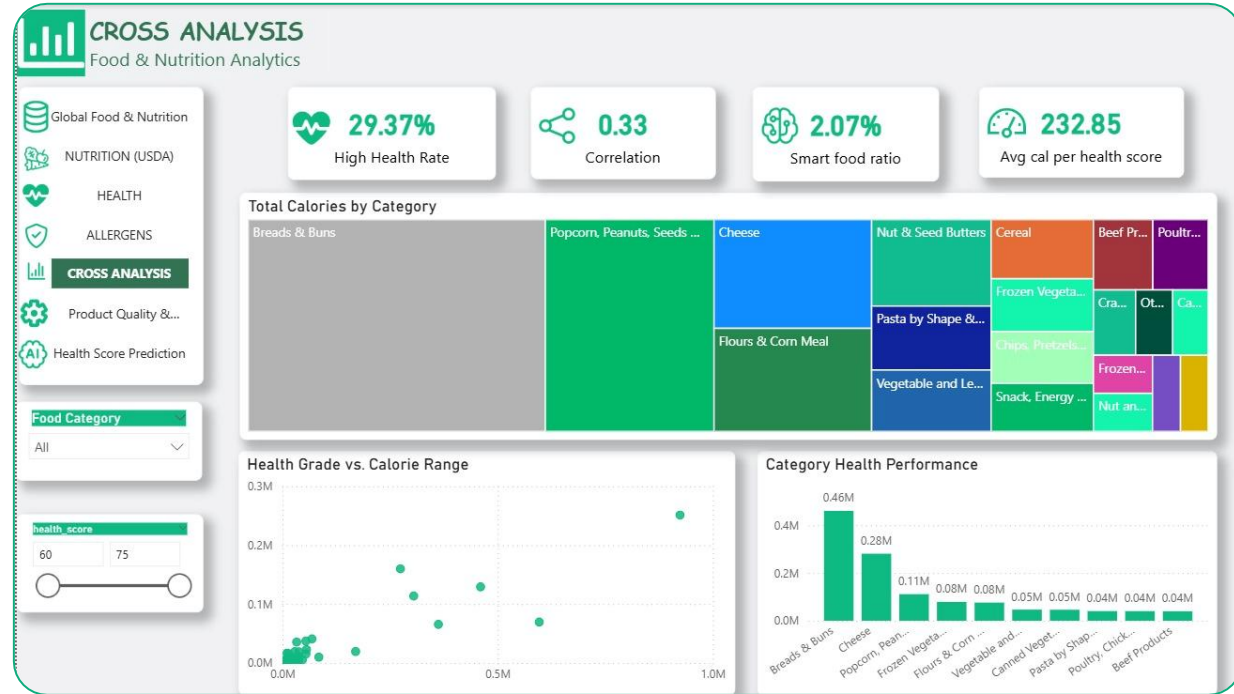
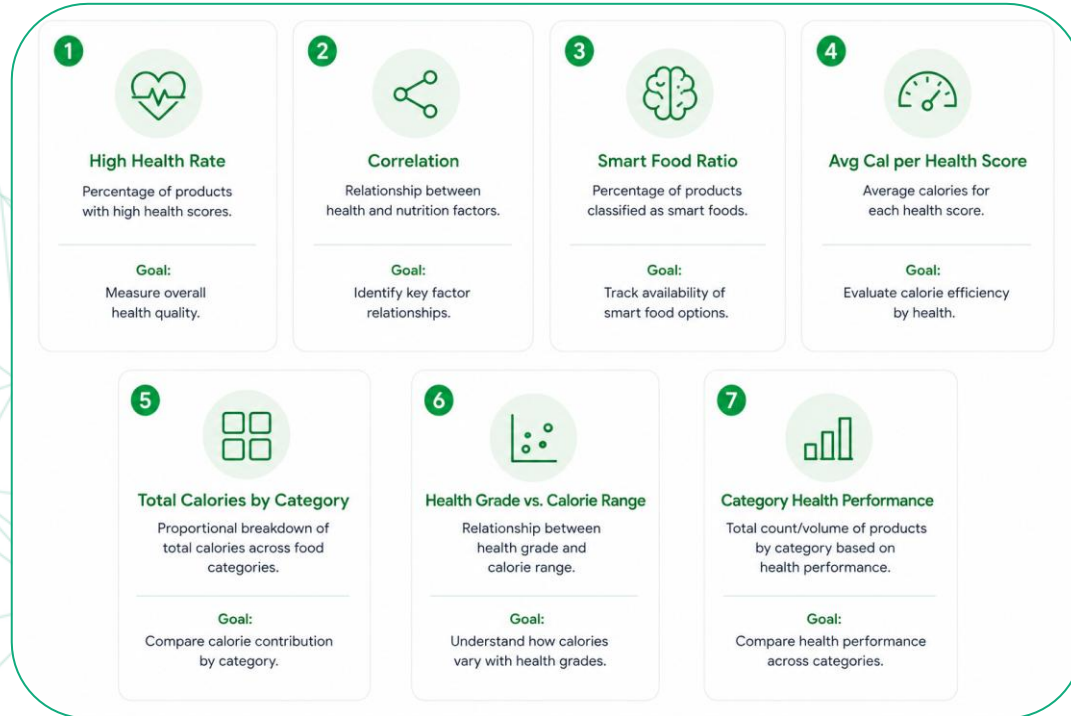
Analyze food allergens and product safety to help users identify safer food choices and reduce allergy risks.





Cross analysis

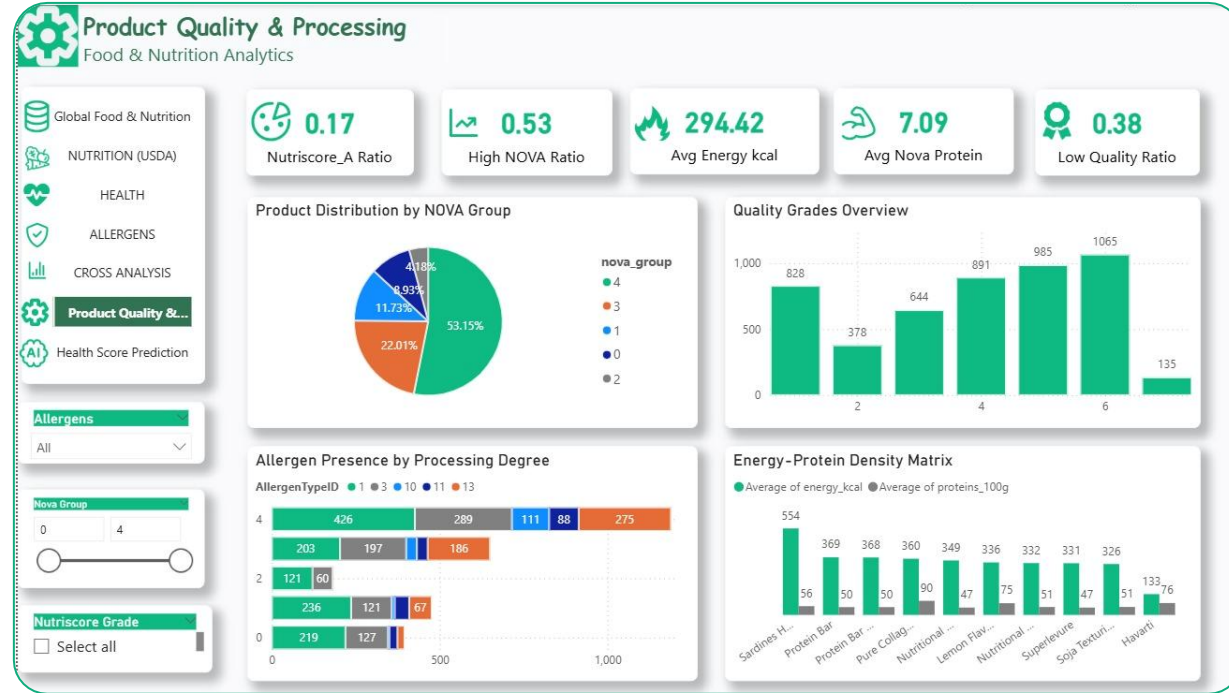
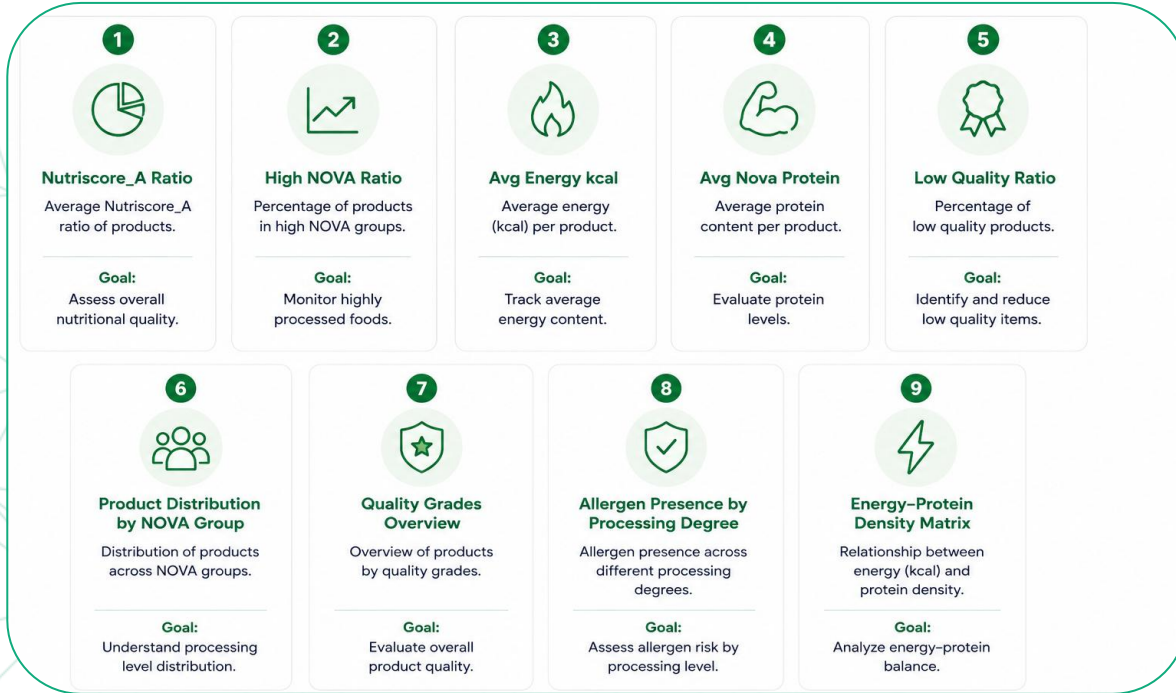
Analyze relationships between nutrition, calories, and health scores to identify healthier food patterns and category performance.





Product Quality & Processing

Analyze product quality, processing levels, and nutritional value to support healthier and safer food choices.





Health Score Prediction

Predict health scores accurately using machine learning.

- 

R² Score

Model prediction accuracy.

Goal: Measure model fit.
- 

MAE

Average prediction error.

Goal: Evaluate error size.
- 

RMSE

Error magnitude score.

Goal: Detect large errors.
- 

Actual vs Predicted

Real vs predicted scores.

Goal: Compare predictions.
- 

Feature Importance

Most impactful features.

Goal: Identify key predictors.
- 

Absolute Error

Prediction error distribution.

Goal: Analyze accuracy gaps.
- 

Absolute Error vs Actual Score

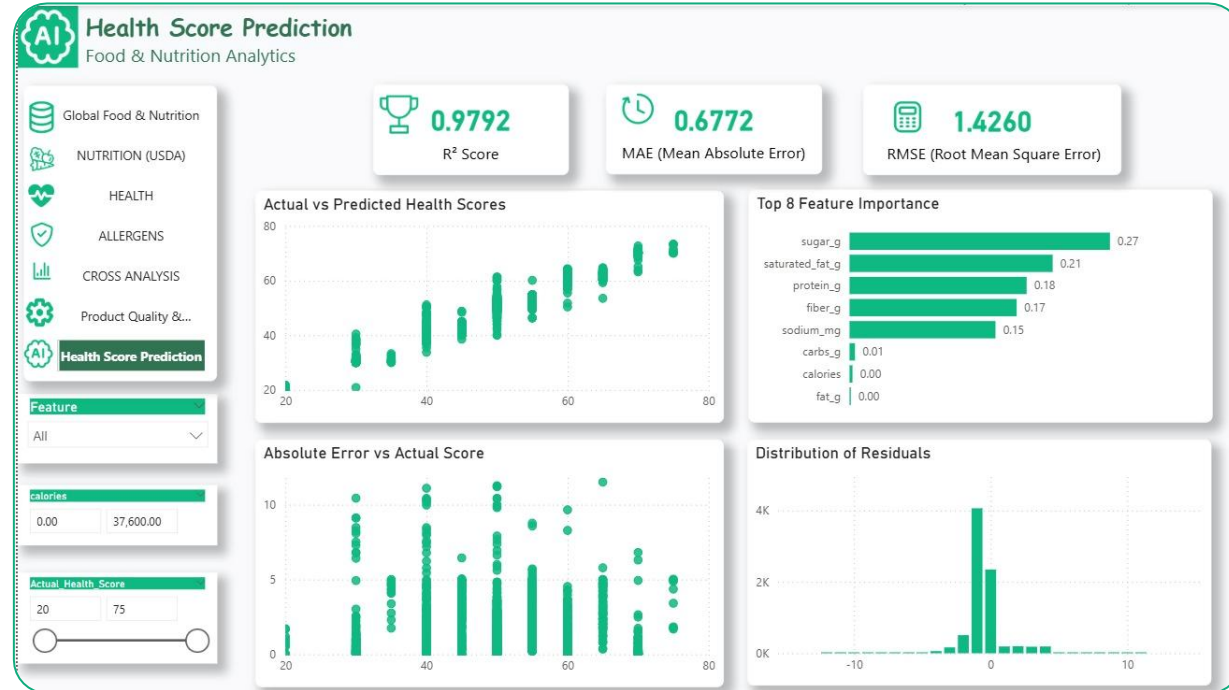
Error size by actual scores.

Goal: Understand error patterns.
- 

Residual Distribution

Spread of residual errors.

Goal: Check model stability.



Project Team + Roles



Mayar and Hend + Alaa

Data Cleaning & Preprocessing

Machine Learning

- Training machine learning models
- Model evaluation
- Predictions and insights



Hend

Power BI Visualization

- Data visualization
- Power BI dashboard creation
- Data insights



Nada

Dashboard Design & Implementation

- Dashboard layout and design
- Dashboard interactivity
- Implementation and testing



Rewida

Documentation & Presentation

- Technical documentation
- Project reports & presentations
- Documentation review

Thank you